

CONSTRUCTION OS

PRD_3

Client Comment Mentions and Task Creation

Full-Stack Developer Specification

Document type	Additive Product Requirements Document
Compatibility	Extends PRD_1 and PRD_2; does not replace existing permissions, timelines, approvals, or task orchestration rules
Primary module	Clients / Sales / Tasks
Version	1.0
Status	Ready for technical review and implementation estimation
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Confidential product specification

Document Control

Relationship to earlier PRDs

PRD_3 is an additive specification. Where this document is silent, PRD_1 and PRD_2 remain authoritative. Existing task history, role-based permissions, department deliverables, approvals, notifications, immutable audit events, and CEO visibility continue to apply.

Item	Definition
Feature name	Client Comment Mentions and Task Creation
Primary user	Sales user
Primary source object	Client profile comment
Primary output	Linked task or cross-department parent task
Mention targets	Department or individual company user
Default behavior	Task creation is suggested when a valid mention is detected
Primary traceability	Permanent two-way link between client comment and task
Out of scope	A comment alone does not constitute technical, legal, financial, or managerial approval

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1. Purpose and Business Outcome

Sales conversations frequently create work for Architecture, Legal, Engineering, Finance, Procurement, Project Management, and other departments. The system must convert a client-facing comment into a controlled, traceable task without forcing the Sales user to leave the client profile or recreate the same information in the Tasks module.

Core principle

A mention is a routing instruction, not an approval. The resulting work must continue through the standard task orchestration, department accountability, approval, file revision, audit, and completion rules defined in PRD_1 and PRD_2.

CCT-001 - Create work from context: A Sales user shall be able to create an actionable task directly from a client comment while preserving the original client context, comment text, attachments, author, and timestamp.

CCT-002 - Avoid duplicate entry: The system shall reuse information already entered in the comment and shall not require the Sales user to type the same request again in a separate task form.

CCT-003 - Preserve traceability: The client comment and generated task shall remain permanently linked in both directions.

CCT-004 - Respect existing control rules: The feature shall not bypass task permissions, department managers, required approvals, decision ownership, document controls, or final completion authority.

2. User Story and Example

2.1 Primary user story

As a Sales user, while reviewing a client profile, I want to mention a department or a specific colleague inside a client comment so that the system creates and routes a formal task to the correct people without losing the client conversation that triggered the request.

2.2 Example client profile

<http://localhost:3100/clients/cm5c8mvhe00033uu7yqs47e8k>

2.3 Example comment

This client needs one bathroom removed. @ArchitectureDepartment @ArchitectName

2.4 Expected result

- The comment is saved on the client profile.
- One linked parent task is created.
- The Architecture Department becomes the accountable department.
- The named architect becomes the initial contributor or assignee.
- The Sales user becomes the task creator and an automatic watcher.
- The task contains the client, project or unit context when available.
- The task and client timelines record the creation and assignment events.

3. Functional Scope

In scope	Out of scope / handled elsewhere
@ mention autocomplete for departments and users	Replacing the existing Task module
Optional task creation from a client comment	Treating a normal comment as formal approval
One parent task with department deliverables	Creating one duplicate task for every department employee
Two-way comment-task linking	Exposing restricted client or contract information
Assignment to department inbox and named contributors	Allowing Sales to approve architectural or legal work
Notifications and status summary on the client page	Replacing department manager accountability

In scope	Out of scope / handled elsewhere
Attachment and metadata transfer	Silently modifying a task after a comment is edited

4. Mention Discovery and Selection

4.1 Trigger

CCT-010 - Mention trigger: Typing the @ character in the client comment editor shall open a searchable autocomplete panel at the cursor position.

CCT-011 - Search behavior: Search shall support partial matching by department name, user name, position, and role. Results must be restricted to the current company and to targets the Sales user is allowed to involve.

4.2 Result groups

Group	Displayed information	Example
Departments	Department name, department icon, manager or coordinator status	Architecture Department
Users	Full name, avatar or initials, job position, department, active status	Elira Hoxha - Architect - Architecture

4.3 Mention token behavior

- A selected mention is inserted as a structured token, not plain text only.
- The token stores the immutable user ID or department ID.
- Renaming a user or department shall update its displayed label without breaking historical references.
- Department and user mentions shall use different icons and accessible labels.
- Duplicate mention tokens shall be prevented before submission.
- Inactive users shall not be selectable for new assignments.

5. Task Creation Logic

5.1 Detection and confirmation

CCT-020 - Task creation prompt: When one or more valid mentions are present, the editor shall show a visible option labeled Create task for mentioned participants.

CCT-021 - Default state: The option should be enabled by default for Sales users, but the Sales user may turn it off and post a notification-only comment.

CCT-022 - No hidden creation: A task shall never be created without a clear visual indication before the comment is submitted.

5.2 Expandable task settings

Field	Requirement	Default
Task title	Required and editable	Generated from the comment where possible
Responsible department	Required when a department is mentioned	Mentioned department
Assigned user	Optional unless only one user is mentioned	Mentioned user
Priority	Required	Normal
Due date	Required by company policy; otherwise optional	Company default SLA
Related project	Optional but recommended	Inherited from client when available
Related unit/property	Optional	Inherited from selected client opportunity
Formal approval required	Boolean or workflow template selector	Based on company routing rules

Field	Requirement	Default
Watchers	Optional	Sales creator automatically included

5.3 Atomic submission

Posting the comment and creating the task must be processed as one controlled transaction or as an idempotent workflow that can recover safely. The user must not end up with a visible comment but an unknown task state.

1. Validate comment content, mentions, permissions, and task settings.
2. Create and save the client comment.
3. Create the task and department deliverables.
4. Create the permanent CommentTaskLink.
5. Copy or reference eligible attachments.
6. Write client and task timeline events.
7. Queue notifications after the transaction is committed.

5.4 Generated task content

Task property	Source
Task ID	System number series
Title	User-edited generated title or Client Request from Sales
Description	Original comment text plus structured source metadata
Source type	CLIENT_COMMENT
Source ID	ClientComment.id
Client	Current client profile
Project / property / unit	Inherited where a valid relationship exists
Creator	Sales user
Responsible departments	Department mentions
Assignees / contributors	User mentions
Attachments	Comment attachments linked or copied according to document policy
Visibility	Calculated using task and client permissions

6. Department and Individual Assignment Rules

6.1 Department-only mention

@ArchitectureDepartment

- Create one department deliverable, not one task per architect.
- Route the deliverable to the Architecture Department inbox.
- Notify the department manager or configured task coordinator.
- Allow the department manager to assign one or more architects.
- Keep the department manager accountable until the deliverable is completed or formally transferred.

6.2 Individual-only mention

@ArchitectName

- Create one task assigned directly to the selected active user.
- Infer the user's department for reporting and routing purposes.
- Notify the user and, where company policy requires, the department manager.
- If the user lacks permission to access the relevant client context, grant only the minimum task-scoped access or block creation and request an authorized assignee.

6.3 Department plus individual

@ArchitectureDepartment @ArchitectName

Assignment result

The department is the accountable owner. The named architect is the initial contributor or assignee. The department manager retains authority to reassign while the original assignment remains in the audit timeline.

6.4 Multiple individuals in one department

Create one department deliverable with multiple contributors unless the user explicitly selects separate tasks. The task manager or department manager defines whether one user is the primary assignee.

7. Multi-Department Orchestration

7.1 Default model

When more than one department is mentioned, the system shall create one parent task and a separate required deliverable for each department. This preserves one client request while allowing independent accountability, deadlines, files, approvals, and completion states.

Mention	Generated deliverable	Typical required output
Architecture Department	Architecture review	Feasibility, revised plan, area impact, drawing package
Legal Department	Legal review	Contract amendment, variation, confirmation, or reason no action is required
Engineering Department	Technical review	MEP/structural impact and technical approval
Finance Department	Commercial approval	Budget impact, approval threshold, payment or pricing confirmation

7.2 Dependency example

Architecture feasibility approved
 -> Engineering impact review
 -> Cost confirmation
 -> Legal variation
 -> Sales communicates approved result to client

The parent task cannot be marked complete while any mandatory deliverable is open, rejected, blocked, overdue without resolution, or awaiting required approval.

7.3 Department response options

- Accept assignment
- Request more information
- Submit comment only
- Upload files or revisions
- Add another department through the standard involvement workflow
- Submit deliverable for approval
- Mark deliverable complete when authorized
- Reject or mark not required with mandatory reason

8. Task and Comment Synchronization

8.1 Client comment display

CCT-040 - Task badge: The source comment shall show the generated task ID, task title, overall status, current responsible department, due date, and overdue state.

CCT-041 - Open linked task: Selecting the task badge shall open the task detail view.

CCT-042 - Status refresh: The client page shall refresh task status through real-time updates or a reliable query invalidation strategy.

8.2 Task source display

CCT-043 - Source reference: The task shall display Created from client comment with the client name, comment author, source date, and a link back to the exact comment anchor.

CCT-044 - Permanent relationship: Deleting or editing the comment shall not delete or silently rewrite the task.

8.3 Comment editing after task creation

- Preserve the original comment snapshot used to create the task.
- Store the updated comment separately as the current comment version.
- Create a task timeline event showing who edited the source comment and when.
- Optionally allow the Sales user to publish the edited text as a task clarification; this must be an explicit action.

8.4 Attachment handling

Attachments must preserve file ID, uploader, timestamp, category, permissions, version, and source relationship. The implementation may use a shared file reference or a controlled copy, but the system must prevent broken links and must not overwrite approved task documents.

9. Permissions and Information Visibility

Role	Allowed	Not allowed
Sales user	Create comment, mention authorized targets, create task, add context/files, watch progress, answer questions	Approve architecture/legal work, bypass manager, close technical deliverable unless separately authorized
Department manager	Accept department work, assign contributors, set department deadline, approve department deliverable	Expose restricted client data to unauthorized users
Mentioned employee	View task-scoped context, comment, upload files, complete assigned actions	See unrelated financial, contract, or client data
Project Manager	Coordinate linked project work according to existing task rules	Override approvals without recorded authority and reason
CEO / authorized executive	View full task history according to PRD_1 rules	Create invisible changes or delete audit events

9.1 Access calculation

Minimum necessary access

A mention does not automatically grant full access to the complete client profile. The recipient receives only the client and project fields required to perform the assigned task unless broader permissions already exist.

- The backend, not the browser, must validate company membership and permissions.
- Cross-company mentions are not allowed unless a JV or external collaboration workflow explicitly permits them.
- Sensitive client notes, prices, contracts, and identity data must remain protected by existing field- and object-level permissions.
- External contractor users are excluded from this feature unless a separate authorized contractor assignment is created.

10. Notifications and Timelines

10.1 Notification recipients

- Mentioned individual users
- Mentioned department manager or task coordinator

- Sales user who created the task
- Relevant Project Manager when a project is linked
- Additional watchers selected in the task panel

10.2 Notification content

New client request from Sales
Client: Maria Rossi
Request: Remove one bathroom
Responsible department: Architecture
Assigned user: John Architect
Due date: 08 August 2026
Task: TSK-2026-00452

10.3 Timeline events

Client timeline	Task timeline
Comment created	Task created from client comment
Department/user mentioned	Source client and comment recorded
Task created and assigned	Department deliverables created
Task status changed	Assignment and reassignment events
Department response submitted	Files, approvals, and decisions
Task completed or reopened	Completion, rejection, or reopening events

11. User Interface Requirements

11.1 Comment editor

- Rich text or controlled plain-text editor with @ mention support.
- Keyboard navigation: Arrow keys, Enter, Escape, Backspace over tokens.
- Mouse and touch selection.
- Department and user grouping.
- Search debounce and loading state.
- Empty, error, inactive-user, and no-permission states.
- Attachment upload before submission.
- Visible Create task toggle when mentions exist.
- Expandable task settings panel.
- Mobile-responsive mention panel that does not cover the active text field.

11.2 Client profile task summary

Displayed item	Interaction
Task ID and title	Open full task
Overall status	Show current workflow state
Department statuses	Open department deliverable
Assignee	Open user profile where allowed
Due date / overdue	Open timeline and delay explanation
Latest response	Open task comments

11.3 Suggested component structure

ClientCommentComposer

- MentionAutocomplete
- MentionToken
- AttachmentPicker
- CreateTaskToggle

- TaskSettingsPanel

ClientCommentCard

- CommentBody
- MentionChips
- LinkedTaskBadge
- LinkedTaskStatusSummary

12. Backend Data Model

12.1 Recommended entities

Entity	Purpose	Key relationships
ClientComment	Stores the source comment	Client, author, attachments, versions
CommentMention	Stores structured department/user mentions	Comment, user or department
CommentTaskLink	Permanent two-way relationship	Comment, task, creation metadata
Task	Parent work object	Client, project, source comment
TaskDepartment	Accountable departments	Task, department, manager
DepartmentDeliverable	Required departmental output	TaskDepartment, assignees, deadline
TaskParticipant	Users and participation type	Task, user, role
TaskEvent	Immutable audit event	Task, actor, timestamp
TaskFile	Task attachment or document link	Task, source attachment, version
TaskNotification	Delivery and read state	Task, recipient, channel

12.2 CommentMention constraints

CommentMention

- id
- companyId
- commentId
- mentionType: USER | DEPARTMENT
- mentionedUserId: nullable
- mentionedDepartmentId: nullable
- createdByUserId
- createdAt
- taskCreated: boolean
- linkedTaskId: nullable

- Exactly one of mentionedUserId or mentionedDepartmentId must be populated.
- The referenced target must belong to the same company unless an approved collaboration rule applies.
- A unique constraint should prevent duplicate mention targets within one comment version.
- Historical mentions shall not be deleted when a user or department is later deactivated.

12.3 CommentTaskLink constraints

CommentTaskLink

- id
- companyId
- clientId
- commentId
- taskId
- createdByUserId
- createdAt
- sourceCommentVersionId
- idempotencyKey

13. API Requirements

13.1 Suggested endpoints

```
GET /api/mentions/search?q=&context=client-comment
POST /api/clients/:clientId/comments
POST /api/clients/:clientId/comments/:commentId/create-task
GET /api/clients/:clientId/comments/:commentId/task
GET /api/departments/:departmentId/available-assignees
PATCH /api/tasks/:taskId/assignment
```

13.2 Preferred combined command

Where practical, use one command endpoint that creates the comment and optional task atomically. If separate endpoints are retained, require an idempotency key and expose an explicit recovery state.

```
POST /api/clients/:clientId/comments

{
  "content": "This client needs one bathroom removed.",
  "mentions": [
    { "type": "DEPARTMENT", "departmentId": "architecture-department-id" },
    { "type": "USER", "userId": "architect-user-id" }
  ],
  "createTask": true,
  "task": {
    "title": "Client Change Request - Remove One Bathroom",
    "priority": "NORMAL",
    "dueDate": "2026-08-08",
    "projectId": "optional-project-id",
    "unitId": "optional-unit-id"
  },
  "idempotencyKey": "uuid-generated-by-client"
}
```

13.3 Response requirements

```
{
  "comment": { "id": "...", "createdAt": "..." },
  "task": { "id": "...", "number": "TSK-2026-00452", "status": "ACTIVE" },
  "link": { "id": "..." },
  "warnings": []
}
```

14. Validation, Idempotency, and Edge Cases

Scenario	Required behavior
Mention removed before submit	Do not store mention or create assignment for that target.
Inactive mentioned user	Block selection or require reassignment before task creation.
Department has no manager	Route to department inbox and notify Company Admin.
User leaves company later	Keep historical record; return open work to department reassignment queue.
Comment posted with createTask off	Save mentions as notification-only; no task is created.
Department and employee duplicated	Create one department deliverable; attach user as contributor.
Multiple departments	Create one parent task with separate deliverables by default.
Client has no project	Create client-only task and permit later project linking.
Comment edited after task creation	Preserve source snapshot; log edit; do not silently update task.

Scenario	Required behavior
Network retry / double click	Use idempotency key; return the original comment/task response.
Permission changes during submission	Revalidate at commit time and fail safely.
Attachment upload partially fails	Do not report full success; show recoverable attachment state.

15. Security and Audit Requirements

CCT-080 - Server-side authorization: Every mention, assignment, client relationship, task creation, and file link must be authorized on the server.

CCT-081 - Tenant isolation: All records and queries must include company or tenant scope. Client-supplied company IDs must not be trusted without session validation.

CCT-082 - Immutable audit: Creation, assignment, reassignment, source edits, attachment changes, permission failures, and task completion must generate immutable audit events.

CCT-083 - Content safety: Comment content and filenames must be sanitized for display. File uploads must pass existing malware, size, type, and storage policies.

CCT-084 - Rate limits: Mention search and notification endpoints must be rate limited and protected from enumeration of users outside the requester's scope.

CCT-085 - No implicit privilege: Mentioning a user or department shall not elevate the Sales user or recipient beyond the minimum permissions required by the task.

16. Acceptance Criteria

AC-01 A Sales user can type @ in a client comment and search authorized departments and active employees.

AC-02 Results visually distinguish departments from individual users.

AC-03 The Sales user can mention Architecture, Legal, or another authorized department.

AC-04 The Sales user can mention one or more specific employees.

AC-05 Posting the comment can create one linked task.

AC-06 A department mention creates one department deliverable, not one task per department employee.

AC-07 A department plus individual mention assigns the department as accountable and the user as contributor or initial assignee.

AC-08 Multiple departments create separate deliverables under one parent task by default.

AC-09 The generated task contains the source comment snapshot, client relationship, attachments, creator, mentions, and project/unit links where available.

AC-10 The comment displays the linked task ID, title, status, assignee, and due date.

AC-11 The task links back to the exact client comment.

AC-12 Mentioned users and department coordinators receive notifications after successful creation.

AC-13 Department managers can reassign work while preserving assignment history.

AC-14 Sales can follow and clarify the task but cannot approve technical or legal work without separate permission.

AC-15 Every important action appears in the client and task timelines.

AC-16 Duplicate mentions and duplicate task creation are prevented.

AC-17 Existing client, task, department, and file permissions remain enforced.

AC-18 The feature works on desktop, tablet, and mobile.

AC-19 Automated tests cover authorization, idempotency, assignment logic, and multi-department creation.

17. Test Scenarios and Implementation Notes

17.1 Minimum end-to-end scenarios

Test	Scenario	Expected result
E2E-01	Sales mentions Architecture Department only	One task; Architecture deliverable routed to department inbox.
E2E-02	Sales mentions one architect only	One task; architect assigned; department inferred.
E2E-03	Sales mentions Architecture plus named architect	Department accountable; architect initial contributor.
E2E-04	Sales mentions Architecture and Legal	One parent task; two department deliverables.
E2E-05	Sales posts comment with task toggle off	Comment and mention notification only; no task.
E2E-06	Request submitted twice with same idempotency key	Only one comment and one task exist.
E2E-07	Mentioned user becomes inactive before commit	Submission is blocked or user removed with explicit warning.
E2E-08	Source comment edited after task creation	Task source snapshot unchanged; edit event recorded.
E2E-09	Unauthorized Sales user searches restricted department	Restricted target is not returned or selectable.
E2E-10	Task completed by departments	Client page reflects final status and completion timeline.

17.2 Technical implementation guidance

- Use structured mention nodes or tokens in the editor payload; do not parse final display text as the source of truth.
- Use a command/service layer for comment-to-task creation so the same logic can later support comments on projects, contracts, meetings, RFIs, and other objects.
- Publish domain events only after the primary database transaction commits.
- Use an outbox pattern or equivalent reliable message mechanism for notifications and search indexing.
- Keep task source metadata generic: sourceObjectType and sourceObjectId, while retaining a strongly validated CommentTaskLink for this feature.
- Avoid copying client-sensitive information into notifications; notifications should contain only safe summary fields and deep links.
- Index departments and users for fast mention search, but always apply authorization filters before returning results.
- Reuse the existing Task orchestration APIs and entities rather than creating a parallel Sales-task implementation.

17.3 Definition of done

Release gate

The feature is ready only when frontend, backend, permission checks, audit events, task orchestration, notifications, responsive behavior, automated tests, and migration scripts are complete and reviewed together. A visual mention component without controlled task creation is not considered complete.

Appendix A - Example Complete Flow

- Sales opens the client profile and writes: This client needs one bathroom removed.
- Sales types @ArchitectureDepartment and @ArchitectName.
- The editor displays the Create task option and pre-fills the task title, department, user, client, and related unit.
- Sales sets priority and due date, attaches the client sketch, and submits.
- The backend validates permissions, creates the comment, task, Architecture deliverable, participants, file links, and timeline events.

13. Architecture manager and named architect receive notifications. Sales becomes a watcher.
14. The architect reviews feasibility and uploads revised PDF and DWG files.
15. If Engineering or Legal is needed, Architecture uses the standard department-involvement workflow defined in PRD_1.
16. After required reviews and approvals, the responsible department completes its deliverable.
17. The client profile displays the final task result, and Sales communicates the approved outcome to the client.

Appendix B - Non-Conflict Rules

- PRD_3 adds a task creation entry point inside client comments; it does not create a second task system.
- Task status, overdue handling, approvals, contract links, files, department deliverables, timelines, and CEO visibility continue to use PRD_1 rules.
- Role Centers, Work Inbox, number series, posting, ledgers, filters, and other NAV-inspired patterns continue to use PRD_2 rules where applicable.
- When a conflict is discovered during implementation, the team must document it and request a product decision rather than silently overriding an earlier rule.